JAVA LEVEL 3

BASIC 1D ARRAY SYNTAX

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i : arr){
            System.out.printf("%d\n",i);
        }
    }
}
```

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        input.nextLine();
        String[] str = new String[n];
        for(int i=0;i<n;i++){
            str[i] = input.nextLine();
        }
        for(String i : str){
            System.out.printf("%s ",i);
        }
    }
}
```

BASIC 2D ARRAY SYNTAX

```
import java.util.*;
```

```
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int r = input.nextInt();
        int c = input.nextInt();
        int[][] arr = new int[r][c];
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                arr[i][j] = input.nextInt();
            }
        }
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                System.out.printf("%d ",arr[i][j]);
            }
            System.out.println();
        }
    }
}
```

1.Sum and average of elements in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int sum=0;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
            sum+=arr[i];
        }
        int avg = sum/n;
        System.out.printf("%d",sum);
        System.out.printf("\n%d",avg);
```

```
    }
}
```

Output

5

5 8 9 6 3

31

6

2.maximum and minimum element in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int max = arr[0];
        int min = arr[0];
        for(int i=0;i<n;i++){
            if(arr[i]>max){
                max=arr[i];
            }
            if(arr[i]<min){
                min=arr[i];
            }
        }
        System.out.printf("Maximum element : %d\n",max);
        System.out.printf("Minimum element : %d",min);
    }
}
```

Output

5

2 52 86 489 21

Maximum element : 489

Minimum element : 2

3.left rotation

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int k = obj.nextInt();
        int temp;
        for(int i=0;i<k;i++){
            temp=arr[0];
            for(int j=0;j<n-1;j++){
                arr[j]=arr[j+1];
            }
            arr[n-1]=temp;
        }
        for(int i : arr){
            System.out.printf("%d ",i);
        }
    }
}
```

Output

5

7 8 9 6 3

2

9 6 3 7 8

4.right rotation

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
```

```
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int k = obj.nextInt();
        int temp;
        for(int i=0;i<k;i++){
            temp = arr[n-1];
            for(int j=n-1;j>=1;j--){
                arr[j]=arr[j-1];
            }
            arr[0]=temp;
        }
        for(int i : arr){
            System.out.printf("%d ",i);
        }
    }
}
```

Output

5

7 8 9 6 3

2

6 3 7 8 9

5.number of occurrence of a number

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int k = obj.nextInt();
```

```
        int count=0;
        for(int i=0;i<n;i++){
            if(arr[i]==k){
                count++;
            }
        }
        System.out.printf("%d",count);
    }
}
```

Output
8
4 8 9 6 8 5 7 6
8
2

6.sum and product of elements in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int sum=0,mul=1;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
            sum+=arr[i];
            mul*=arr[i];
        }
        System.out.printf("%d\n",sum);
        System.out.printf("%d",mul);
    }
}
```

Output
6
4 8 5 2 3 1

23
960
7.square of array elements
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
            s = arr[i] * arr[i];
            System.out.printf("%d ",s);
        }
    }
}
```
Output
5
4 2 6 3 2
16 4 36 9 4
8.difference between maximum and minimum number
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int max=arr[0],min=arr[0],diff;
        for(int i=0;i<n;i++){
            if(arr[i]>max){
```

```
            max=arr[i];
        }
        if(arr[i]<min){
            min=arr[i];
        }
    }
    System.out.printf("%d %d",max,min);
    System.out.printf("\n%d",diff = max - min);
  }
}
```

Output

5

28 96 52 41 23

96 23

73

9.remove first occurrence of an element in an array

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner obj = new Scanner(System.in);
    int n = obj.nextInt();
    int s;
    int[] arr = new int[n];
    for(int i=0;i<n;i++){
        arr[i] = obj.nextInt();
    }
    int k = obj.nextInt(),first=1;
    for(int i=0;i<n;i++){
        if(arr[i]==k && first==1){
            first=0;
            continue;
        }
        System.out.printf("%d ",arr[i]);
    }
  }
```

```
}
```
Output
7
2 4 6 3 2 8 3
3
2 4 6 2 8 3
10.removing last occurred duplicates
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int count;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]==arr[j]){
                    for(int k=j;k<n-1;k++){
                        arr[j]=arr[j+1];
                    }
                    n--;
                }
            }
        }
        for(int i=0;i<n;i++){
            System.out.printf("%d ", arr[i]);
        }
    }
}
```
Output
6

3 8 4 3 10 8
3 8 4 10
11.unique elements
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int count;
        for(int i=0;i<n;i++){
            count=0;
            for(int j=0;j<n;j++){
                if(arr[i]==arr[j] && i!=j){
                    count++;
                }
            }
            if(count==0){
                System.out.printf("%d ",arr[i]);
            }
        }
    }
}
```
Output
7
3 5 2 7 3 8 2
5 7 8
12.removing element according to index in a array
```
import java.util.*;
public class Main{
    public static void main(String[] args){
```

```
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int k = obj.nextInt();
        for(int i=0;i<n;i++){
            if(i==k){
                continue;
            }
            System.out.printf("%d ",arr[i]);
        }
    }
}
```

Output

8

23 45 76 453 675 21 987 564

5

23 45 76 453 675 987 564

13.remove element based on position in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int s;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int k = obj.nextInt();
        for(int i=0;i<n;i++){
            if(i==k-1){
```

```
            for(int j=i;j<n-1;j++){
                arr[j]=arr[j+1];
            }
            n--;
        }
    }
    for(int i=0;i<n;i++){
        System.out.printf("%d ", arr[i]);
    }
  }
}
```

Output

5

7 9 6 3 2

4

7 9 6 2

14.non prime numbers

```
import java.util.*;
public class Main{
  public static void main(String args[]){
    Scanner obj = new Scanner(System.in);
    int n = obj.nextInt();
    int[] arr = new int[n];
    for(int i=0;i<n;i++){
        arr[i] = obj.nextInt();
    }
    int prime;
    for(int i=0;i<n;i++){
        prime=1;
        for(int j=2;j<arr[i];j++){
            if(arr[i]%j==0){
                prime=0;
            }
        }
        if(prime==0){
```

```
            System.out.printf("%d ",arr[i]);
        }
    }
  }
}
```

Output

8

7 6 3 22 5 86 20 13

6 22 86 20

15.prime numbers

```
import java.util.*;
public class Main{
  public static void main(String args[]){
    Scanner obj = new Scanner(System.in);
    int n = obj.nextInt();
    int[] arr = new int[n];
    for(int i=0;i<n;i++){
      arr[i] = obj.nextInt();
    }
    int prime;
    for(int i=0;i<n;i++){
      prime=1;
      for(int j=2;j<arr[i];j++){
        if(arr[i]%j==0){
          prime=0;
        }
      }
      if(prime==1){
        System.out.printf("%d ",arr[i]);
      }
    }
  }
}
```

Output

8

7 6 3 22 5 86 20 13
7 3 5 13
16.sum of first,second,last and last before number

```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int sum=arr[0]+arr[1]+arr[n-2]+arr[n-1];
        System.out.printf("%d",sum);
    }
}
```

Output
6
2 4 5 1 3 6
15
17.second largest element

```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int firstlargest = arr[0],secondlargest=arr[1];
        for(int i=0;i<n;i++){
            if(arr[i]>firstlargest){
                secondlargest = firstlargest;
                firstlargest = arr[i];
```

```
            }
            else if(arr[i]>secondlargest && arr[i]<firstlargest){
                secondlargest = arr[i];
            }
        }
        System.out.printf("%d",secondlargest);
    }
}
```

Output

6

77 96 85 12 525 425

425

18.ascending order

```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int temp;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    temp=arr[j];
                    arr[j]=arr[i];
                    arr[i]=temp;
                }
            }
        }
        for(int i=0;i<n;i++){
            System.out.printf("%d ",arr[i]);
        }
```

```
    }
}
```
Output
6
12 63 85 96 41 56
12 41 56 63 85 96
19.descending order
```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int temp;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]<arr[j]){
                    temp=arr[j];
                    arr[j]=arr[i];
                    arr[i]=temp;
                }
            }
        }
        for(int i=0;i<n;i++){
            System.out.printf("%d ",arr[i]);
        }
    }
}
```
Output
6
12 63 85 96 41 56
96 85 63 56 41 12

20.chunksize

```
import java.util.*;
public class Main{
    public static void main(String args[]){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int chunksize = obj.nextInt();
        for(int i=0;i<n;i+=chunksize){
            System.out.print("[");
            for(int j=i;j<i+chunksize && j<n;j++){
                System.out.printf("%d",arr[j]);
                if(j<i+chunksize-1 && j<n-1){
                    System.out.print(",");
                }
            }
            System.out.printf("]");
        }
    }
}
```

Output

5

10 52 85 63 41

2

[10,52][85,63][41]

21.insert element according to position in array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n+1];
```

```
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int pos = obj.nextInt();
        int v = obj.nextInt();
        for(int i=n;i>=pos;i--){
            arr[i] = arr[i-1];
        }
        arr[pos - 1] = v;
        for(int i=0;i<=n;i++){
            System.out.printf("%d ",arr[i]);
        }
    }
}
```

Output

```
5
1 4 7 8 5
3
2
1 4 2 7 8 5
```

22.Find missing letter in a sequence of consecutive letters

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        char[] arr = new char[n];
        for(int i=0;i<n;i++){
            arr[i] = input.next().charAt(0);
        }
        System.out.print("Original elements in array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(" , ");
```

```
            }
        }
        System.out.print("]\nMissing element : ");
        for(int i=0;i<n;i++){
            if(arr[i+1] != arr[i]+1){
                System.out.printf("%c",arr[i]+1);
                break;
            }
        }
    }
}
```

Output

```
6
p q s t u v
Original elements in array : [p , q , s , t , u , v]
Missing element : r
```

23.Check if array is negative dominant

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int count1=0,count2=0;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        System.out.print("Array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]\n");
```

```
        for(int i=0;i<n;i++){
            if(arr[i] < 0){
                count1++;
            }
            if(arr[i] > 0){
                count2++;
            }
        }
        if(count1 > count2){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

6

1 2 5 -4 3 -6

Array : [1, 2, 5, -4, 3, -6]

False

24.alternate positive and negative elements in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        System.out.print("Array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(", ");
```

```
            }
        }
        System.out.print("]\n");
        int isTrue=1;
        for(int i=0;i<n-1;i++){
            if(!((arr[i]>0 && arr[i+1]<0) || (arr[i]<0 && arr[i+1]>0))){
                isTrue = 0;
            }
        }
        if(isTrue == 1){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

Output

5

-1 2 3 4 -5

Array : [-1, 2, 3, 4, -5]

False

25.Check if numbers in array can form a consecutive list

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        System.out.print("Array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
```

```
            if(i < n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]\n");
        int temp,isTrue = 1;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    temp = arr[j];
                    arr[j] = arr[i];
                    arr[i] = temp;
                }
            }
        }
        for(int i=0;i<n-1;i++){
            if(arr[i+1] != arr[i] + 1){
                isTrue = 0;
                break;
            }
        }
        if(isTrue == 1){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

Output

6

20 21 19 24 25 22

Array : [20, 21, 19, 24, 25, 22]

False

24.largest gap between sorted elements in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        System.out.print("Array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]\n");
        int temp;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    temp = arr[j];
                    arr[j] = arr[i];
                    arr[i] = temp;
                }
            }
        }
        int max = arr[0];
        for(int i=0;i<n-1;i++){
            if(arr[i+1] - arr[i] > max){
                max = arr[i+1] - arr[i];
            }
        }
        System.out.print(max);
    }
```

```
}
```

Output

7

23 -2 45 38 12 4 6

Array : [23, -2, 45, 38, 12, 4, 6]

15

25.find triplets summing to a target value in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        System.out.print("Array : [");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]\n");
        int temp;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    temp = arr[j];
                    arr[j] = arr[i];
                    arr[i] = temp;
                }
            }
        }
```

```
        for(int i=0;i<n-2;i++){
            if(i>0 && arr[i] == arr[i-1]) continue;
            for(int j=i+1;j<n-1;j++){
                if(j>i+1 && arr[j] == arr[j-1]) continue;
                for(int z=j+1;z<n;z++){
                    if(z>j+1 && arr[z] == arr[z-1]) continue;
                    if(arr[i]+arr[j]+arr[z] == target){
                        System.out.printf("%d %d %d\n",arr[i],arr[j],arr[z]);
                    }
                }
            }
        }
    }
}
```

Output

```
8
1 6 3 0 8 4 1 7
7
Array : [1, 6, 3, 0, 8, 4, 1, 7]
0 1 6
0 3 4
```

26.largest number from a list of non negative numbers

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        String[] arr = new String[n];
        int[] nums = new int[n];
        for(int i=0;i<n;i++){
            nums[i] = input.nextInt();
        }
        for(int i =0;i<n;i++){
            arr[i] = String.valueOf(nums[i]);
        }
```

```
        for(int i=0;i<n-1;i++){
            for(int j=i+1;j<n;j++){
                String temp;
                String s1 = arr[i] + arr[j];
                String s2 = arr[j] + arr[i];
                if(s1.compareTo(s2)<0){
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        if(arr[0] == "0"){
            System.out.print("0");
            return;
        }
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
        }
    }
}
```

Output

6

1 2 3 0 4 6

643210

27.find continuous subarray to sort for array to be sorted

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] nums = new int[n];
        int a=0,b=0;
        for(int i=0;i<n;i++){
```

```
            arr[i] = input.nextInt();
            nums[i] = arr[i];
        }
        int temp;
        for(int i=0;i<n;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    temp = arr[j];
                    arr[j] = arr[i];
                    arr[i] = temp;
                }
            }
        }
        for(int i=0;i<n;i++){
            if(arr[i] != nums[i]){
                a = i;
                break;
            }
        }
        for(int i=n-1;i>=0;i--){
            if(arr[i] != nums[i]){
                b = i;
                break;
            }
        }
        System.out.print("Contiguous subarray : \n");
        for(int i=a;i<=b;i++){
            System.out.print(nums[i]+" ");
        }
    }
}
```

Another code:

```
import java.util.*;
public class Main {
```

```
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] sorted = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = input.nextInt();
            sorted[i] = arr[i];
        }
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                if (sorted[i] > sorted[j]) {
                    int temp = sorted[i];
                    sorted[i] = sorted[j];
                    sorted[j] = temp;
                }
            }
        }
        int start = 0, end = n - 1;
        while (start < n && arr[start] == sorted[start]) {
            start++;
        }
        while (end > start && arr[end] == sorted[end]) {
            end--;
        }
        System.out.println("Continuous subarray:");
        for (int i = start; i <= end; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

Output

8

1 2 6 5 3 8 4 9

Continuous subarray:

6 5 3 8 4

28.subarray of an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n;i++){
            for(int j=i;j<n;j++){
                for(int k=i;k<=j;k++){
                    System.out.print(arr[k] + " ");
                }
                System.out.print("\n");
            }
        }
    }
}
```

Output

```
6
5 2 1 0 3 6
5
5 2
5 2 1
5 2 1 0
5 2 1 0 3
5 2 1 0 3 6
2
2 1
2 1 0
2 1 0 3
2 1 0 3 6
```

1
1 0
1 0 3
1 0 3 6
0
0 3
0 3 6
3
3 6
6

29.Find smallest length subarray with sum >= specified value

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int start=-1,end=-1,len,minlen=n+1;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        for(int i=0;i<n;i++){
            int sum=0;
            for(int j=i;j<n;j++){
                sum += arr[j];
                if(sum>=target){
                    len = j-i+1;
                    if(len<minlen){
                        minlen = len;
                    }
                break;
                }
            }
        }
```

```
        System.out.print(minlen);
    }
}
```

Output

5

1 2 3 4 6

8

2

30.Find subarray with sum >= specified value

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int start=-1,end=-1,len,minlen=n+1;
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        for(int i=0;i<n;i++){
            int sum=0;
            for(int j=i;j<n;j++){
                sum += arr[j];
                if(sum >= target){
                    len = j-i+1;
                    if(len<minlen){
                        minlen = len;
                        start = i;
                        end = j;
                    }
                    break;
                }
            }
        }
```

```
        for(int i=start;i<=end;i++){
            System.out.print(arr[i]+" ");
        }
    }
}
```

Output

5

1 2 3 4 6

8

4 6

31.Find subarray with largest sum in a given array

```
import java.util.*;
public class Main{
    public static void main(String[]  args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int max=Integer.MIN_VALUE,sum=0;
        for(int i=0;i<n;i++){
            for(int j=i;j<n;j++){
                sum=0;
                for(int k=i;k<=j;k++){
                    sum += arr[k];
                }
                if(sum>max){
                    max = sum;
                }
            }
        }
        System.out.print(max);
    }
}
```

Output
5
5 4 1 7 8
25

32.Find maximum difference between two elements in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int min = arr[0];
        int maxDiff = Integer.MIN_VALUE;
        for(int i=1;i<n;i++){
            if(arr[i] - min > maxDiff){
                maxDiff = arr[i] - min;
            }
            if(arr[i] < min){
                min = arr[i];
            }
        }
        System.out.print(maxDiff);
    }
}
```

Output
9
2 3 1 7 9 5 11 3 5
10

33.Replace each element with product of other elements

```
import java.util.*;
```

```
public class Main{
    public static void main(String[]  args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int mul;
        for(int i=0;i<n;i++){
            mul = 1;
            for(int j=0;j<n;j++){
                if(i==j) continue;
                mul *= arr[j];
            }
            nums[i] = mul;
        }
        System.out.print("[");
        for(int i=0;i<n;i++){
            System.out.print(nums[i]);
            if(i<n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
    }
}
```

Output

7

1 2 3 4 5 6 7

[5040, 2520, 1680, 1260, 1008, 840, 720]

34.find equilibrium indices in an array

```
import java.util.*;
public class Main{
```

```
    public static void main(String[]  args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n;i++){
            int left = 0,right=0;
            for(int j=0;j<i;j++){
                left += arr[j];
            }
            for(int k=i+1;k<n;k++){
                right += arr[k];
            }
            if(left == right){
                System.out.print(i + " ");
            }
        }
    }
}
```

Output

7

-7 1 5 2 -4 3 0

3 6

35.Rearrange array with every second element greater

```
import java.util.*;
public class Main{
    public static void main(String[]  args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
```

```
        }
        int temp;
        for(int i=0;i<n-1;i++){
            if(i%2==0){
                if(arr[i] > arr[i+1]){
                    temp = arr[i];
                    arr[i] = arr[i+1];
                    arr[i+1] = temp;
                }
            }
            else{
                if(arr[i] < arr[i+1]){
                    temp = arr[i];
                    arr[i] = arr[i+1];
                    arr[i+1] = temp;
                }
            }
        }
        System.out.print("[");
        for(int i=0;i<n;i++){
            System.out.print(arr[i]);
            if(i < n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
    }
}
```

Output

```
11
1 2 4 9 5 3 8 7 10 12 14
[1, 4, 2, 9, 3, 8, 5, 10, 7, 14, 12]
```

36.Find maximum product of two integers in an array

```
import java.util.*;
```

```
public class Main{
    public static void main(String[]  args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int a=0,b=0;
        int mul,max=Integer.MIN_VALUE;
        for(int i=0;i<n;i++){
            for(int j=0;j<n;j++){
                if(i==j) continue;
                mul = arr[i] * arr[j];
                if(mul>max){
                    max=mul;
                    a = arr[i];
                    b = arr[j];
                }
            }
        }
        System.out.printf("Pair : [%d,%d]\n",a,b);
        System.out.printf("Maximum product : %d",max);
    }
}
```

Output

```
8
2 3 5 7 -7 5 8 -5
Pair : [7,8]
Maximum product : 56
```

37.print all subarray with sum 0 in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
```

```
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int sum=0;
        System.out.print("Subarray with sum 0 :\n");
        for(int i=0;i<n;i++){
            sum=0;
            for(int j=i;j<n;j++){
                sum += arr[j];
                if(sum == 0){
                    System.out.printf("Subarray [%d,%d] : ",i,j);
                    for(int k=i;k<=j;k++){
                        System.out.print(arr[k]+" ");
                    }
                    System.out.print("\n");
                }
            }
        }
    }
}
```

Output

```
10
1 3 -7 3 2 3 1 -3 -2 -2
Subarray with sum 0 :
Subarray [0,3] : 1 3 -7 3
Subarray [1,8] : 3 -7 3 2 3 1 -3 -2
```

38.Check if array contains a subarray with sum zero

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
```

```
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int sum=0,isTrue=0;
        for(int i=0;i<n;i++){
            sum=0;
            for(int j=i;j<n;j++){
                sum += arr[j];
                if(sum == 0){
                    isTrue = 1;
                    break;
                }
            }
            break;
        }
        if(isTrue==1){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

Output

6

1 2 -3 4 5 6

True

39.Replace each element with the next greatest element

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
```

```
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] nums = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int max = Integer.MIN_VALUE;
        for(int i=0;i<n;i++){
            max = Integer.MIN_VALUE;
            for(int j=i+1;j<n;j++){
                if(arr[j]>max){
                    max = arr[j];
                }
            }
            nums[i] = max;
        }
        nums[n-1] = -1;
        System.out.print("[");
        for(int i=0;i<n;i++){
            System.out.print(nums[i]);
            if(i<n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
    }
}
```

Output

7

45 20 100 23 -5 2 -6

[100, 100, 23, 2, 2, -6, -1]

40.Separate even and odd numbers in an array

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] even = new int[n];
        int[] odd = new int[n];
        int index1=0,index2=0;
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n;i++){
            if(arr[i]%2==0){
                even[index1++] = arr[i];
            }
        }
        for(int i=0;i<n;i++){
            if(arr[i]%2 != 0){
                odd[index2++] = arr[i];
            }
        }
        for(int i=0;i<index1;i++){
            System.out.print(even[i]+" ");
        }
        for(int i=0;i<index2;i++){
            System.out.print(odd[i]+" ");
        }
    }
}
```

Output

10

1 2 3 4 5 6 7 8 9 10

2 4 6 8 10 1 3 5 7 9

41.Separate 0s on left and 1s on right in array

```
import java.util.*;
```

```
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        System.out.print("[");
        for(int i=0;i<n;i++){
            if(arr[i] == 0){
                System.out.print(arr[i]);
                if(i<n-1){
                    System.out.print(", ");
                }
            }
        }
        System.out.print(", ");
        for(int i=0;i<n;i++){
            if(arr[i] == 1){
                System.out.print(arr[i]);
                if(i<n-2){
                    System.out.print(", ");
                }
            }
        }
        System.out.print("]");
    }
}
```

Output

```
9
0 0 1 1 0 1 1 1 0
[0, 0, 0, 0, 1, 1, 1, 1, 1]
```

42.Sort array with alternate max-min values

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] nums = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        Arrays.sort(arr);
        int right = n - 1;
        int left = 0;
        System.out.print("[");
        for(int i=0;i<n;i++){
            if(i%2==0){
                nums[i] = arr[right--];
            }
            else{
                nums[i] = arr[left++];
            }
        }
        for(int i=0;i<n;i++){
            System.out.print(nums[i]);
            if(i<n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
    }
}
```

Output

10

10 30 20 60 50 40 90 80 100 70

[100, 10, 90, 20, 80, 30, 70, 40, 60, 50]

43.array elements with positive,zero,negative order

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int printed = 0;
        System.out.print("[");
        for(int i=0;i<n;i++){
            if(arr[i]>0){
                System.out.print(arr[i]);
                printed++;
                if(printed < n){
                    System.out.print(", ");
                }
            }
        }
        for(int i=0;i<n;i++){
            if(arr[i]==0){
                System.out.print(arr[i]);
                printed++;
                if(printed < n){
                    System.out.print(", ");
                }
            }
        }
        for(int i=0;i<n;i++){
            if(arr[i]<0){
                System.out.print(arr[i]);
```

```
                printed++;
                if(printed < n){
                    System.out.print(", ");
                }
            }
        }
        System.out.print("]");
    }
}
```

Output

10

-4 8 6 -5 6 -2 1 2 3 -11

[8, 6, 6, 1, 2, 3, -4, -5, -2, -11]

44.Removing vowels in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int len = str.length();
        for(int i=0;i<len;i++){
            char c = str.charAt(i);
            if(c == 'a' || c=='e' || c=='i' || c=='o' || c=='u' || c=='A' || c=='E' || c=='I'
|| c=='O' || c=='U'){
                continue;
            }
            else{
                System.out.printf("%c",c);
            }
        }
    }
}
```

Output

HELLO world

HLL wrld

45.unicode value of a character

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int n = obj.nextInt();
        System.out.print((int)str.charAt(n));
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int n = obj.nextInt();
        System.out.printf("%d",Character.codePointAt(str,n));
    }
}
```

Output

hello

1

101

46.compare two strings lexicographically

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str1 = obj.nextLine();
        String str2 = obj.nextLine();
        int result = str1.compareTo(str2);
```

```
        if(result==0){
            System.out.printf("%s and %s are equal",str1,str2);
        }
        else if(result<0){
            System.out.printf("\"%s\" is smaller than \"%s\"",str1,str2);
        }
        else{
            System.out.printf("\"%s\" is greater than \"%s\"",str1,str2);
        }
    }
}
```

Output

apple

Apple

"apple" is greater than "Apple"

47.compare two strings lexicographically ignoring case difference

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str1 = obj.nextLine();
        String str2 = obj.nextLine();
        int result = str1.compareToIgnoreCase(str2);
        if(result==0){
            System.out.printf("%s and %s are equal",str1,str2);
        }
        else if(result<0){
            System.out.printf("\"%s\" is smaller than \"%s\"",str1,str2);
        }
        else{
            System.out.printf("\"%s\" is greater than \"%s\"",str1,str2);
        }
    }
}
```

Output
apple
AppLE
apple and AppLE are equal

48.concatenate two strings

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str1 = obj.nextLine();
        String str2 = obj.nextLine();
        String result = str1.concat(str2);
        System.out.print(result);
    }
}
```

Output
hello
world
hello world

49.specified word is present in the string or not

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str1 = obj.nextLine();
        String str2 = obj.nextLine();
        String[] tokens = str1.split("\\s+");
        int len = tokens.length;
        for(int i=0;i<len;i++){
            if(tokens[i].equals(str2)){
                System.out.print("True");
                return;
            }
```

```
        }
        System.out.print("False");
    }
}
```

Output
the sky is blue
and
False

50.given string ends with another string.

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s1 = obj.nextLine();
        String str = obj.nextLine();
        if(s1.endsWith(str)){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

Output
apple
es
False

51.substring from a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
```

```
        int len = str.length();
        int begin = obj.nextInt();
        int end = obj.nextInt();
        if(begin < len && end<=len){
            System.out.print(str.substring(begin,end));
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int len = str.length();
        int begin = obj.nextInt();
        int end = obj.nextInt();
        if(begin < len && end<=len){
            for(int i=begin;i<end;i++){
                System.out.print(str.charAt(i));
            }
        }
    }
}
```

Output

```
helloworld
3 9
loworl
```

52.Check If Letters of One String Exist in Another

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s1 = obj.nextLine();
```

```
        String s2 = obj.nextLine();
        if(s1.contains(s2)){
            System.out.print("True");
        }
        else{
            System.out.print("False");
        }
    }
}
```

Output

helloworld

low

True

53.remove specified word from string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s1 = obj.nextLine();
        String s2 = obj.nextLine();
        String[] tokens = s1.trim().split("\\s+");
        int len = tokens.length;
        for(int i=0;i<len;i++){
            if(tokens[i].equals(s2)){
                continue;
            }
            System.out.print(tokens[i]+" ");
        }
    }
}
```

Output

the sky is blue

is

the sky blue

54.frequency of all characters in string
In alphabetical order

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        final int ASCII = 256;
        String str = obj.nextLine();
        int len = str.length();
        int[] freq = new int[ASCII];
        for(int i=0;i<len;i++){
            freq[str.charAt(i)]++;
        }
        for(int i=0;i<ASCII;i++){
            if(freq[i] != 0){
                System.out.printf("%c : %d\n",(char)i,freq[i]);
            }
        }
    }
}
```

Output

```
the sky is blue
  : 3
b : 1
e : 2
h : 1
i : 1
k : 1
l : 1
s : 2
t : 1
u : 1
y : 1
```

Without alphabetical order

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        final int ASCII = 256;
        String str = obj.nextLine();
        int len = str.length();
        int[] freq = new int[ASCII];
        for(int i=0;i<len;i++){
            freq[str.charAt(i)]++;
        }
        for(int i=0;i<len;i++){
            char c = str.charAt(i);
            if(freq[c] != 0){
                System.out.printf("%c : %d\n",c,freq[c]);
                freq[c]=0;
            }
        }
    }
}
```

Output

```
the sky is blue
t : 1
h : 1
e : 2
  : 3
s : 2
k : 1
y : 1
i : 1
b : 1
l : 1
u : 1
```

55.Count Duplicates Occurring More Than Twice

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        final int ASCII = 256;
        String str = obj.nextLine();
        int len = str.length();
        int[] freq = new int[ASCII];
        int count=0;
        for(int i=0;i<len;i++){
            freq[str.charAt(i)]++;
        }
        for(int i=0;i<ASCII;i++){
            if(freq[i] > 2){
                count++;
            }
        }
        System.out.print(count);
    }
}
```

Output

create a date calender

3

56.Reverse Odd-Length Words

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        for(int i=0;i<n;i++){
            int len = tokens[i].length();
```

```
            if(len%2 != 0){
                for(int j=len-1;j>=0;j--){
                    System.out.print(tokens[i].charAt(j));
                }
            }
            else{
                System.out.print(tokens[i]);
            }
            System.out.print(" ");
        }
    }
}
```

Output

It is hot but raining today

It is toh tub gniniar yadot

57.Reverse even-Length Words

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        for(int i=0;i<n;i++){
            int len = tokens[i].length();
            if(len%2 == 0){
                for(int j=len-1;j>=0;j--){
                    System.out.print(tokens[i].charAt(j));
                }
            }
            else{
                System.out.print(tokens[i]);
            }
            System.out.print(" ");
```

```
        }
    }
}
```

Output

the sky is blue

the sky si eulb

58.reverse first and last word in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        for(int i=0;i<n;i++){
            if(i==0 || i==n-1){
                int len = tokens[i].length();
                for(int j=len-1;j>=0;j--){
                    System.out.print(tokens[i].charAt(j));
                }
            }
            else{
                System.out.print(tokens[i]);
            }
            System.out.print(" ");
        }
    }
}
```

Output

it is HOT and RAINING today

ti is HOT and RAINING yadot

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        for(int i=0;i<n;i++){
            if(i==0 || i==n-1){
                int len = tokens[i].length();
                for(int j=len-1;j>=0;j--){
                    System.out.print(tokens[i].charAt(j));
                }
                System.out.print(" ");
            }
        }
    }
}
```

Output

the sky is blue

eht eulb

59.reverse words in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        for(int i=0;i<n;i++){
            int len = tokens[i].length();
            for(int j=len-1;j>=0;j--){
                System.out.print(tokens[i].charAt(j));
            }
            System.out.print(" ");
        }
```

```
    }
}
```
Output
Tomorrow is Saturday
worromoT si yadrutaS

60. Swap First and Last Elements

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int temp1 = arr[0];
        int temp2 = arr[n-1];
        arr[0] = temp2;
        arr[n-1] = temp1;
        for(int i=0;i<n;i++){
            System.out.print(arr[i]+" ");
        }
    }
}
```
Output
6
50 10 90 40 70 60
60 10 90 40 70 50

61.Merge two sorted arrays maintaining order

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
```

```
int n = obj.nextInt();
int[] arr1 = new int[n];
for(int i=0;i<n;i++){
    arr1[i] = obj.nextInt();
}
int m = obj.nextInt();
int[] arr2 = new int[m];
for(int i=0;i<m;i++){
    arr2[i] = obj.nextInt();
}
int a = m+n;
int[] arr = new int[a];
for(int i=0;i<n;i++){
    arr[i] = arr1[i];
}
for(int i=0;i<m;i++){
    arr[i+n] = arr2[i];
}
Arrays.sort(arr);
System.out.print("A : [");
for(int i=0;i<n;i++){
    System.out.print(arr[i]);
    if(i<n-1){
        System.out.print(", ");
    }
}
System.out.print("]\n");
System.out.print("A : [");
for(int i=n;i<a;i++){
    System.out.print(arr[i]);
    if(i<a-1){
        System.out.print(", ");
    }
}
System.out.print("]\n");
```

```
    }
}
```

Output

```
6
1 5 6 7 8 10
3
2 4 9
A : [1, 2, 4, 5, 6, 7]
B : [8, 9, 10]
```

62.print the first negative number in the subarray of size n from the given array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        int z = obj.nextInt();
        for(int i=0;i<n;i++){
            for(int j=i;j<n;j++){
                int count=0,first=1,a=0;
                for(int k=i;k<=j;k++){
                    count++;
                    if(arr[k]<0 && first==1){
                        a=arr[k];
                        first=0;
                    }
                }
                if(count==z){
                    if(a != 0){
                        System.out.print(a+" ");
```

```
                }
                else{
                    System.out.print("0");
                }
            }
        }
    }
  }
}
```

Output

1) 10
10 -5 6 8 -3 7 9 -2 4 -1
3
-5 -5 -3 -3 -3 -2 -2 -2

2) 5
1 2 3 4 5
2
0 0 0 0

63.Longest Prefix and Suffix Substring

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int len = str.length();
        int maxlen=0;
        for(int i=0;i<len;i++){
            boolean isMatch = true;
            for(int j=0;j<i;j++){
                if(str.charAt(j) != str.charAt(len-i+j)){
                    isMatch = false;
                    break;
```

```
            }
        }
        if(isMatch){
            maxlen = i;
        }
    }
    if(maxlen>0){
        for(int i=0;i<maxlen;i++){
            System.out.print(str.charAt(i));
        }
    }
    else{
        System.out.print("No longest string found");
    }
  }
}
```

Output

playersplay

play

64.Remove Char Except First and Last

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner obj = new Scanner(System.in);
    String str = obj.nextLine();
    char c = obj.next().charAt(0);
    int n = str.length();
    for(int i=0;i<n;i++){
        if(str.charAt(i)==c && i!=0 && i!=n-1){
            continue;
        }
        else{
            System.out.print(str.charAt(i));
        }
```

```
        }
    }
}
```

Output

zebrazone

z

zebraone

65.Print all leader elements in array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        for(int i=0;i<n;i++){
            boolean isTrue = true;
            for(int j=i+1;j<n;j++){
                if(arr[i]<arr[j]){
                    isTrue = false;
                }
            }
            if(isTrue){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

Output

7

10 9 14 23 15 0 9

23 15 9

66.Find majority element in array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] counted = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
            counted[i]=0;
        }
        int count=0;
        int m = n/2;
        for(int i=0;i<n;i++){
            if(counted[i]==0){
                counted[i]=1;
                count=1;
                for(int j=i+1;j<n;j++){
                    if(arr[i]==arr[j]){
                        count++;
                        counted[j]=1;
                    }
                }
                if(count>m){
                    System.out.print(arr[i]);
                }
            }
        }
    }
}
```

Output

15

1 6 6 5 7 4 1 7 7 7 7 7 7 7 2

67.Frequency of each element in array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        int[] counted = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
            counted[i]=0;
        }
        int count=0;
        for(int i=0;i<n;i++){
            if(counted[i]==0){
                count=1;
                for(int j=i+1;j<n;j++){
                    if(arr[i]==arr[j]){
                        count++;
                        counted[j]=1;
                    }
                }
                System.out.printf("%d : %d\n",arr[i],count);
            }
        }
    }
}
```

Output

15

1 6 6 5 7 4 1 7 7 7 7 7 7 7 2

1 : 2

6 : 2

5 : 1

7 : 8
4 : 1
2 : 1

68.reverse vowels in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        char[] c = str.toCharArray();
        int n = c.length;
        int i=0,j=n-1;
        while(i<j){
            while(i<j && !(c[i]=='a' || c[i]=='e' || c[i]=='i' || c[i]=='o' || c[i]=='u' ||
c[i]=='A' || c[i]=='E' || c[i]=='I' || c[i]=='O' || c[i]=='U')){
                i++;
            }
            while(i<j && !(c[j]=='a' || c[j]=='e' || c[j]=='i' || c[j]=='o' || c[j]=='u' ||
c[j]=='A' || c[j]=='E' || c[j]=='I' || c[j]=='O' || c[j]=='U')){
                j--;
            }
            char temp = c[i];
            c[i]=c[j];
            c[j]=temp;
            i++;
            j--;
        }
        for(int k=0;k<n;k++){
            System.out.print(c[k]);
        }
    }
}
```

Another method

```
import java.util.*;
```

```
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        char[] c = str.toCharArray();
        int n = c.length;
        int i=0,j=n-1;
        String vowels = "aeiouAEIOU";
        while(i<j){
            while(i<j && vowels.indexOf(c[i])==-1){
                i++;
            }
            while(i<j && vowels.indexOf(c[j])==-1){
                j--;
            }
            char temp = c[i];
            c[i]=c[j];
            c[j]=temp;
            i++;
            j--;
        }
        for(int k=0;k<n;k++){
            System.out.print(c[k]);
        }
    }
}
```

Output

a quick brown fox fell into the box

o qeock briwn fex foll onti thu bax

69.reverse consonants in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
```

```
        String str = input.nextLine();
        char[] c = str.toCharArray();
        int n = c.length;
        int i=0,j=n-1;
        while(i<j){
            while(i<j && (!((c[i]>='a' && c[i]<='z') || (c[i]>='A' && c[i]<='Z')) ||
c[i]=='a' || c[i]=='e' || c[i]=='i' || c[i]=='o' || c[i]=='u' || c[i]=='A' || c[i]=='E' ||
c[i]=='I' || c[i]=='O' || c[i]=='U')){
                i++;
            }
            while(i<j && (!((c[j]>='a' && c[j]<='z') || (c[j]>='A' && c[j]<='Z')) ||
c[j]=='a' || c[j]=='e' || c[j]=='i' || c[j]=='o' || c[j]=='u' || c[j]=='A' || c[j]=='E' ||
c[j]=='I' || c[j]=='O' || c[j]=='U')){
                j--;
            }
            char temp = c[i];
            c[i]=c[j];
            c[j]=temp;
            i++;
            j--;
        }
        for(int k=0;k<n;k++){
            System.out.print(c[k]);
        }
    }
}
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        char[] c = str.toCharArray();
        int n = c.length;
        int i=0,j=n-1;
        String vowels = "aeiouAEIOU";
```

```
        while(i<j){
            while(i<j && (!((c[i]>='a' && c[i]<='z') || (c[i]>='A' && c[i]<='Z')) ||
vowels.indexOf(c[i])!=-1)){
                i++;
            }
            while(i<j && (!((c[j]>='a' && c[j]<='z') || (c[j]>='A' && c[j]<='Z')) ||
vowels.indexOf(c[j])!=-1)){
                j--;
            }
            char temp = c[i];
            c[i]=c[j];
            c[j]=temp;
            i++;
            j--;
        }
        for(int k=0;k<n;k++){
            System.out.print(c[k]);
        }
    }
}
```

Output

a quick brown fox fell into the box

a xuibh ttonl lof xefn iwro bke coq

70.sort strings by length

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        input.nextLine();
        String[] s = new String[n];
        for(int i=0;i<n;i++){
            s[i] = input.nextLine();
        }
```

```
        Arrays.sort(s,(a,b) -> a.length() - b.length());
        System.out.print("Ascending order : \n[");
        for(int j=0;j<n;j++){
            System.out.print(s[j]);
            if(j<n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
        Arrays.sort(s,(a,b) -> b.length() - a.length());
        System.out.print("\nDescending order : \n[");
        for(int k=0;k<n;k++){
            System.out.print(s[k]);
            if(k<n-1){
                System.out.print(", ");
            }
        }
        System.out.print("]");
    }
}
```

Output

```
9
Green
White
Black
Pink
Orange
Blue
Champagne
Indigo
Ivory
Ascending order :
[Pink, Blue, Green, White, Black, Ivory, Orange, Indigo, Champagne]
Descending order :
[Champagne, Orange, Indigo, Green, White, Black, Ivory, Pink, Blue]
```

Another method

Ascending order

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        String[] arr = new String[n];
        for(int i=0;i<n;i++){
            arr[i] = sc.nextLine();
        }
        for(int i=0;i<n-1;i++){
            if(arr[i].length()>arr[i+1].length()){
                String temp = arr[i];
                arr[i] = arr[i+1];
                arr[i+1] = temp;
            }
        }
        for(String j : arr){
            System.out.print(j+" ");
        }
    }
}
```

Output

```
5
green
pink
yellow
blue
orange
pink green blue yellow orange
```

Descending order

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        String[] arr = new String[n];
        for(int i=0;i<n;i++){
            arr[i] = sc.nextLine();
        }
        for(int i=0;i<n-1;i++){
            for(int j=0;j<n-1-i;j++){
                if(arr[j].length()<arr[j+1].length()){
                    String temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }
        for(String i : arr){
            System.out.print(i+" ");
        }
    }
}
```

Output

5
pink
orange
red
green
blue
orange green pink blue red

71.find triplets summing to a given number
Without sorting

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        boolean foundfirst = false;
        System.out.print("[");
        for(int i=0;i<n-2;i++){
            for(int j=i+1;j<n-1;j++){
                for(int k=j+1;k<n;k++){
                    if(arr[i]+arr[j]+arr[k] == target){
                        if(foundfirst){
                            System.out.print(", ");
                        }
                        System.out.printf("[%d, %d, %d]",arr[i],arr[j],arr[k]);
                        foundfirst = true;
                    }
                }
            }
        }
        System.out.print("]");
    }
}
```

Output

6

1 -2 0 5 -1 -4

2

[[1, 5, -4], [-2, 5, -1]]

With sorting

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        Arrays.sort(arr);
        boolean foundfirst = false;
        System.out.print("[");
        for(int i=0;i<n-2;i++){
            if(i>0 && arr[i]==arr[i-1]) continue;
            for(int j=i+1;j<n-1;j++){
                if(j>i+1 && arr[j]==arr[j-1]) continue;
                for(int k=j+1;k<n;k++){
                    if(k>j+1 && arr[k]==arr[k-1]) continue;
                    if(arr[i]+arr[j]+arr[k] == target){
                        if(foundfirst){
                            System.out.print(", ");
                        }
                        System.out.printf("[%d, %d, %d]",arr[i],arr[j],arr[k]);
                        foundfirst = true;
                    }
                }
            }
        }
        System.out.print("]");
    }
}
```

Output

```
6
1 -2 0 5 -1 -4
2
[[-4, 1, 5], [-2, -1, 5]]
```

72.Sum of Digits in String

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        int n = s.length();
        int sum = 0;
        for(int i=0;i<n;i++){
            if(s.charAt(i)>='0' && s.charAt(i)<='9'){
                sum += s.charAt(i) - '0';
            }
        }
        System.out.print(sum);
    }
}
```

Output

ab5c2d4ef12s

14

73.Longest Mirror Image Substring

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        int n = s.length();
        String mirror = "";
        for(int i=0;i<n;i++){
            if(s.charAt(i)==s.charAt(n-1-i)){
                mirror += s.charAt(i);
            }
            else{
                break;
```

```
            }
        }
        System.out.print(mirror);
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        int n = s.length();
        StringBuilder sb = new StringBuilder();
        for(int i=0;i<n;i++){
            if(s.charAt(i)==s.charAt(n-1-i)){
                sb.append(s.charAt(i));
            }
            else{
                break;
            }
        }
        System.out.print(sb.toString());
    }
}
```

Output

```
rotavator
rotavator
```

74.Remove All Instances of Substring

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        int n = s.length();
```

```
        String target = input.nextLine();
        String result = s.replace(target,"");
        System.out.print(result);
    }
}
```

Output

This is the test string

st

This is the te ring

75.Count Words Ending with m or n

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        String[] tokens = s.trim().split("\\s+");
        int n = tokens.length;
        int count = 0;
        for(int i=0;i<n;i++){
            String w = tokens[i];
            int len = w.length();
            if(w.charAt(len-1)=='m' || w.charAt(len-1)=='n'){
                count++;
            }
        }
        System.out.print(count);
    }
}
```

Output

mam is in the room

3

76. Add two same-size matrices

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int r = input.nextInt();
        int c = input.nextInt();
        int[][] m1 = new int[r][c];
        int[][] m2 = new int[r][c];
        int[][] sum = new int[r][c];
        System.out.print("Matrix 1 : \n");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                m1[i][j] = input.nextInt();
            }
        }
        System.out.print("Matrix 2 : \n");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                m2[i][j] = input.nextInt();
            }
        }
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                sum[i][j] = m1[i][j] + m2[i][j];
            }
        }
        System.out.print("Sum : \n");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                System.out.print(sum[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

Output

3 3

```
Matrix 1 :
1 2 3
4 5 6
7 8 9
Matrix 2 :
10 11 12
13 14 15
16 17 18
Sum :
11 13 15
17 19 21
23 25 27
```

77.remove first occurrence of duplicate elements

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int count;
        for(int i=0;i<n;i++){
            count = 0;
            for(int j=i+1;j<n;j++){
                if(arr[i]==arr[j]){
                    count++;
                }
            }
            if(count==0){
                System.out.print(arr[i]+" ");
            }
        }
```

```
    }
}
```

Output

6

10 22 10 20 11 22

10 20 11 22

78.unique elements

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int count;
        for(int i=0;i<n;i++){
            count=0;
            for(int j=0;j<n;j++){
                if(arr[i]==arr[j] && i!=j){
                    count++;
                }
            }
            if(count==0){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

Output

6

10 22 10 20 11 22

20 11

79.palindrome or not

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s = obj.nextLine();
        int n = s.length();
        int i=0,j=n-1;
        boolean isPalindrome = true;
        while(i<j){
            if(s.charAt(i)!=s.charAt(j)){
                isPalindrome = false;
                break;
            }
            i++;
            j--;
        }
        if(isPalindrome){
            System.out.print("Palindrome");
        }
        else{
            System.out.print("Not palindrome");
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
```

```
        StringBuilder sb = new StringBuilder();
        for(int i=n-1;i>=0;i--){
            char c = str.charAt(i);
            sb.append(c);
        }
        String s = sb.toString();
        if(s.equals(str)){
            System.out.print("Palindrome");
        }
        else{
            System.out.print("Not Palindrome");
        }
    }
}
```

Output
malayalam
Palindrome

80.longest palindromic substring

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s = obj.nextLine();
        int n = s.length();
        int maxlen=1,start=0;
        for(int i=0;i<n;i++){
            int left=i,right=i;
            while(left>=0 && right<n && s.charAt(left)==s.charAt(right)){
                left--;
                right++;
            }
            int len1 = right-left-1;
            left=i;
            right = i+1;
```

```
            while(left>=0 && right<n && s.charAt(left)==s.charAt(right)){
                left--;
                right++;
            }
            int len2 = right-left-1;
            int len = Math.max(len1,len2);
            if(len>maxlen){
                maxlen=len;
                start = i - (len-1) / 2;
            }
        }
        for(int i=start;i<start+maxlen;i++){
            System.out.print(s.charAt(i));
        }
    }
}
```

Output

mississippi

ississi

81.longest and shortest word in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String s = obj.nextLine();
        String[] tokens = s.trim().split("\\s+");
        int n = tokens.length;
        String short_word="",long_word="";
        int shortlen = Integer.MAX_VALUE;
        int maxlen = Integer.MIN_VALUE;
        for(int i=0;i<n;i++){
            String word = tokens[i];
            int len = word.length();
            if(len<shortlen){
```

```
            shortlen = len;
            short_word = word;
        }
        if(len>maxlen){
            maxlen = len;
            long_word = word;
        }
    }
    System.out.printf("Longest word : %s\n",long_word);
    System.out.printf("Shortest word : %s",short_word);
  }
}
```

Output

learn java programming

Longest word : programming

Shortest word : java

82.Greatest common divisor of elements in a array

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner obj = new Scanner(System.in);
    int n = obj.nextInt();
    int[] nums = new int[n];
    for(int i=0;i<n;i++){
        nums[i] = obj.nextInt();
    }
    int gcd = nums[0];
    for(int i=0;i<n;i++){
        int a = gcd;
        int b = nums[i];
        while(b!=0){
            int temp = b;
            b = a%b;
            a = temp;
```

```
            }
            gcd = a;
        }
        System.out.printf("GCD : %d",gcd);
    }
}
```

Output

5

2 4 6 8 10

GCD : 2

83.second largest digit in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s = input.nextLine();
        int n = s.length();
        int[] digit = new int[100];
        int count=0;
        for(int i=0;i<n;i++){
            char c = s.charAt(i);
            if(c>='0' && c<='9'){
                digit[count++] = c - '0';
            }
        }
        if(count==0){
            System.out.print("-1");
            return;
        }
        int[] nums = new int[count];
        for(int i=0;i<count;i++){
            nums[i] = digit[i];
        }
        Arrays.sort(nums);
```

```
        int largest = nums[count-1];
        for(int i=count-2;i>=0;i--){
            if(nums[i]!=largest){
                System.out.print(nums[i]);
                return;
            }
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        StringBuilder sb = new StringBuilder();
        int n = s.length();
        for(int i=0;i<n;i++){
            char c = s.charAt(i);
            if(Character.isDigit(c)){
                sb.append(c);
            }
        }
        char[] arr = sb.toString().toCharArray();
        Arrays.sort(arr);
        if(arr.length==0){
            System.out.print("-1");
            return;
        }
        int len = arr.length;
        char max = arr[len-1];
        for(int i=len-2;i>=0;i--){
            if(arr[i]!=max){
                System.out.print(arr[i]);
                return;
```

```
            }
        }
        System.out.print("-1");
    }
}
```

Output

fed1235321dfe

3

helloworld

-1

84.Count Characters in String

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        int count=0,count1=0,count2=0,count3=0;
        for(int i=0;i<n;i++){
            if(str.charAt(i)>='a' && str.charAt(i)<='z' || str.charAt(i)>='A' &&
str.charAt(i)<='Z'){
                count1++;
            }
            else if(str.charAt(i)==' '){
                count2++;
            }
            else if(str.charAt(i)>='0' && str.charAt(i)<='9'){
                count3++;
            }
            else{
                count++;
            }
        }
```

```
        System.out.printf("Letters : %d\nNumbers : %d\nSpace : %d\nOther :
%d\n",count1,count3,count2,count);
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int count1=0,count2=0,count3=0,count4=0;
        for(int i=0;i<s.length();i++){
            char c = s.charAt(i);
            if(Character.isLetter(c)){
                count1++;
            }
            else if(Character.isDigit(c)){
                count2++;
            }
            else if(c==' '){
                count3++;
            }
            else{
                count4++;
            }
        }
        System.out.printf("Letters : %d\nNumnbers : %d\nSpace : %d\nOther :
%d",count1,count2,count3,count4);
    }
}
```

Output

```
Aa kiu, I swd skieo 236587. GH kiu: sieo?? 25.33
Letters : 23
Numbers : 10
Space : 9
```

Other : 6

85.unique three digit numbers

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int len=0,count=0,digits=0;
        int[] arr = new int[10];
        int[] nums = new int[10];
        while(n>0){
            int rev = n%10;
            arr[digits++] = rev;
            n/=10;
        }
        for(int i=digits-1;i>=0;i--){
            nums[len++] = arr[i];
        }
        for(int i=0;i<len;i++){
            for(int j=0;j<len;j++){
                for(int k=0;k<len;k++){
                    if(i!=j && i!=k && j!=k){
                        System.out.printf("%d%d%d\n",nums[i],nums[j],nums[k]);
                        count++;
                    }
                }
            }
        }
        System.out.printf("Count : %d",count);
    }
}
```

Output

1234

123

```
124
132
134
142
143
213
214
231
234
241
243
312
314
321
324
341
342
412
413
421
423
431
432
Count : 24
```

86.capitalize each word

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        for(int i=0;i<n;i++){
            char c = str.charAt(i);
            if(i==0 || str.charAt(i-1)==' '){
```

```
            if(c>='a' && c<='z'){
                c = Character.toUpperCase(c);
            }
        }
        System.out.print(c);
    }
  }
}
```

Output

the sky is blue

The Sky Is Blue

the quick brown fox jumps over the lazy dog.

The Quick Brown Fox Jumps Over The Lazy Dog.

87.print prime numbers from even index in array

```
import java.util.*;
public class Main{
   public static void main(String[] args){
      Scanner input = new Scanner(System.in);
      int n = input.nextInt();
      int[] arr = new int[n];
      for(int i=0;i<n;i++){
         arr[i] = input.nextInt();
      }
      for(int i=0;i<n;i+=2){
         int num = arr[i];
         boolean isPrime = true;
         if(num<=1){
            isPrime = false;
         }
         else{
            for(int j=2;j<=num/2;j++){
               if(num%j==0){
                  isPrime = false;
                  break;
```

```
                }
            }
        }
        if(isPrime){
            System.out.print(num+" ");
        }
    }
  }
}
```

Output

9

2 1 5 3 6 7 4 10 11

2 5 11

88.get a string and find frequency of each word

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        boolean[] visited = new boolean[n];
        for(int i=0;i<n;i++){
            if(visited[i]) continue;
            int count=1;
            for(int j=i+1;j<n;j++){
                if(tokens[i].equals(tokens[j])){
                    count++;
                    visited[j] = true;
                }
            }
            System.out.printf("%s : %d\n",tokens[i],count);
        }
    }
```

```
}
```
Output
apple orange grape fruit banana orange grape apple
apple : 2
orange : 2
grape : 2
fruit : 1
banana : 1

89.sum of array elements except the smallest and the greatest

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] nums = new int[n];
        for(int i=0;i<n;i++){
            nums[i] = input.nextInt();
        }
        int sum=0;
        Arrays.sort(nums);
        for(int i=1;i<n-1;i++){
            sum += nums[i];
        }
        System.out.print(sum);
    }
}
```
Output
6
2 4 1 5 6 3
14

90.get words separately and convert it into sentence with fullstop(.) at the end

```
import java.util.*;
```

```
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        String[] str = new String[n];
        for(int i=0;i<n;i++){
            str[i] = sc.nextLine();
        }
        for(int i=0;i<n;i++){
            String word = str[i];
            if(i==0){
                word =
String.valueOf(Character.toUpperCase(word.charAt(0)))+word.substring(1);
            }
            System.out.print(word);
            if(i<n-1){
                System.out.print(" ");
            }
        }
        System.out.print(".");
    }
}
```

Output

```
4
sun
is
shining
brightly
Sun is shining brightly.
```

91.camel case to snake case

```
import java.util.*;
public class Main{
    public static void main(String[] args){
```

```
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        StringBuilder s = new StringBuilder();
        for(int i=0;i<n;i++){
            char c = str.charAt(i);
            if(Character.isUpperCase(c)){
                if(i!=0){
                    s.append("_");
                }
                s.append(Character.toLowerCase(c));
            }
            else{
                s.append(c);
            }
        }
        System.out.print(s.toString());
    }
}
```

Output

theSkyIsBlue

the_sky_is_blue

92.left rotation in string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = input.nextInt();
        char[] arr = str.toCharArray();
        int len = arr.length;
        for(int i=0;i<n;i++){
            char temp = arr[0];
            for(int j=0;j<len-1;j++){
```

```
            arr[j] = arr[j+1];
        }
        arr[len-1] = temp;
    }
    System.out.print(new String(arr));
  }
}
```

Output
hello
3
lohel

93.right rotation in string

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner input = new Scanner(System.in);
    String str = input.nextLine();
    int n = input.nextInt();
    char[] arr = str.toCharArray();
    int len = arr.length;
    for(int i=0;i<n;i++){
        char temp = arr[len-1];
        for(int j=len-1;j>=1;j--){
            arr[j] = arr[j-1];
        }
        arr[0] = temp;
    }
    System.out.print(new String(arr));
  }
}
```

Output
programming
4
mingprogram

94.print first unique characters index in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int n = str.length();
        boolean[] visited = new boolean[n];
        for(int i=0;i<n;i++){
            if(visited[i]) continue;
            int count=1;
            for(int j=i+1;j<n;j++){
                if(str.charAt(i)==str.charAt(j)){
                    count++;
                    visited[j] = true;
                }
            }
            visited[i] = true;
            if(count==1){
                System.out.print(i);
                break;
            }
        }
    }
}
```

Output

malayalam

4

doodle

4

95.print first unique character in a string

```
import java.util.*;
public class Main{
```

```
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int n = str.length();
        boolean[] visited = new boolean[n];
        for(int i=0;i<n;i++){
            if(visited[i]) continue;
            int count=1;
            for(int j=i+1;j<n;j++){
                if(str.charAt(i)==str.charAt(j)){
                    count++;
                    visited[j] = true;
                }
            }
            visited[i] = true;
            if(count==1){
                System.out.print(str.charAt(i));
                break;
            }
        }
    }
}
```

Output

malayalam

y

96.

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
```

```
        for(int i=0;i<n-1;i++){
            int a=arr[i],b=arr[i+1];
            int num = (int)Math.pow(a,b);
            System.out.print(num+" ");
        }
        System.out.print((int)Math.pow(arr[n-1],arr[0]));
    }
}
```

Output

5

1 2 3 4 5

1 8 81 1024 5

```
import java.util.*;
import java.math.BigInteger;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = obj.nextInt();
        }
        for(int i=0;i<n-1;i++){
            BigInteger base = BigInteger.valueOf(arr[i]);
            int exponent = arr[i+1];
            BigInteger result = base.pow(exponent);
            System.out.print(result+" ");
        }
        BigInteger lastbase = BigInteger.valueOf(arr[n-1]);
        int lastexponent = arr[0];
        BigInteger lastresult = lastbase.pow(lastexponent);
        System.out.print(lastresult);
    }
}
```

Output
5
5 6 7 8 9 10
15625 279936 5764801 134217728 59049

97.run length encoding

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        int i = 0;
        while(i<n){
            char currentChar = str.charAt(i);
            int count=1;
            while(i+1<n && str.charAt(i+1)==currentChar){
                count++;
                i++;
            }
            System.out.print(currentChar+""+count);
            i++;
        }
    }
}
```

Output
abababab
a1b1a1b1a1b1a1b1
aaaabbb
a4b3

98.highest digit that appears consecutively three times in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
```

```
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        char maxdigit = ' ';
        int i=0;
        while(i<n){
            char current = str.charAt(i);
            int count=1;
            while(i+1<n && str.charAt(i+1)==current){
                count++;
                i++;
            }
            if(count>=3 && (maxdigit==' ' || maxdigit < current)){
                maxdigit = current;
            }
            i++;
        }
        if(maxdigit!=' '){
            System.out.print(""+maxdigit+maxdigit+maxdigit);
        }
        else{
            System.out.print("");
        }
    }
}
```

Output

9988877555222

888

99.articles count

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
```

```
        String[] tokens = str.trim().split("\\s+");
        int count = 0;
        for(String word : tokens){
            word = word.replaceAll("[^a-zA-Z]","").toLowerCase();
            if(word.equals("a") || word.equals("an") || word.equals("the")){
                count++;
            }
        }
        System.out.print(count);
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        String[] tokens = str.trim().split("\\s+");
        int n = tokens.length;
        int count=0;
        for(int i=0;i<n;i++){
            String cleaned ="";
            String word = tokens[i];
            for(int j=0;j<word.length();j++){
                char c = word.charAt(j);
                if((c>='a' && c<='z') || (c>='A' && c<='Z')){
                    cleaned += c;
                }
            }
            if(cleaned.equals("a") || cleaned.equals("an") ||
cleaned.equals("the")){
                count++;
            }
        }
        System.out.print(count);
```

```
    }
}
```
Output
an apple a day keeps the doctor away
3

100.square of odd elements
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n;i++){
            if(arr[i]%2!=0){
                int mul = arr[i] * arr[i];
                System.out.print(mul+" ");
            }
        }
    }
}
```
Output
5
1 2 3 4 5
1 9 25

101.keyboard row
```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
```

```
        input.nextLine();
        String[] s = new String[n];
        for(int i=0;i<n;i++){
            s[i] = input.nextLine();
        }
        String s1 = "qwertyuiop";
        String s2 = "asdfghjkl";
        String s3 = "zxcvbnm";
        for(int i=0;i<n;i++){
            String word = s[i];
            String lower = word.toLowerCase();
            boolean r1=true,r2=true,r3=true;
            for(int j=0;j<lower.length();j++){
                char c = lower.charAt(j);
                if(s1.indexOf(c) == -1) r1 = false;
                if(s2.indexOf(c) == -1) r2 = false;
                if(s3.indexOf(c) == -1) r3 = false;
            }
            if(r1 || r2 || r3){
                System.out.print(word+" ");
            }
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        input.nextLine();
        String[] s = new String[n];
        for(int i=0;i<n;i++){
            s[i] = input.nextLine();
        }
```

```
        String s1 = "qwertyuiop";
        String s2 = "asdfghjkl";
        String s3 = "zxcvbnm";
        for(int i=0;i<n;i++){
            String word = s[i];
            String lower = word.toLowerCase();
            boolean r1=true,r2=true,r3=true;
            for(int j=0;j<lower.length();j++){
                char c = lower.charAt(j);
                if(!(s1.contains(c+""))) r1=false;
                if(!(s2.contains(c+""))) r2=false;
                if(!(s3.contains(c+""))) r3=false;
            }
            if(r1 || r2 || r3){
                System.out.print(word+" ");
            }
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        input.nextLine();
        String[] s = new String[n];
        for(int i=0;i<n;i++){
            s[i] = input.nextLine();
        }
        String s1 = "qwertyuiop";
        String s2 = "asdfghjkl";
        String s3 = "zxcvbnm";
        for(int i=0;i<n;i++){
            String word = s[i];
```

```
            String lower = word.toLowerCase();
            boolean r1=true,r2=true,r3=true;
            for(int j=0;j<lower.length();j++){
                char c = lower.charAt(j);
                if(!(s1.contains(String.valueOf(c)))) r1=false;
                if(!(s2.contains(String.valueOf(c)))) r2=false;
                if(!(s3.contains(String.valueOf(c)))) r3=false;
            }
            if(r1 || r2 || r3){
                System.out.print(word+" ");
            }
        }
    }
}
```

Output

```
4
alaska
Peace
HeLLo
dAd
alaska dAd
```

102.floor and ceil

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int target = input.nextInt();
        int floor = -1,ceil = -1;
        for(int i=0;i<n;i++){
```

```
            if(arr[i]<target){
                if(floor == -1 || arr[i]>floor){
                    floor = arr[i];
                }
            }
            if(arr[i]>target){
                if(ceil == -1 || arr[i]<ceil){
                    ceil = arr[i];
                }
            }
        }
        System.out.print(floor+" "+ceil);
    }
}
```

Output

6

30 20 22 15 25 10

25

22 30

103.differentiate vowels and consonants

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        String s1 = "aeiou";
        String s2 = "bcdfghjklmnpqrstvwxyz";
        String lower = str.toLowerCase();
        for(int i=0;i<n;i++){
            char c = lower.charAt(i);
            if(s1.contains(c+"")){
                System.out.printf("%c is a vowel\n",c);
            }
```

```
            if(s2.contains(c+"")){
                System.out.printf("%c is a consonant\n",c);
            }
        }
    }
}
```

Output

Hello World
h is a consonant
e is a vowel
l is a consonant
l is a consonant
o is a vowel
w is a consonant
o is a vowel
r is a consonant
l is a consonant
d is a consonant

104.apple distribution in boxes(using arrays)

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int box = input.nextInt();
        int a = n/box;
        int rem = n%box;
        int[] distributed = new int[box];
        for(int i=0;i<box;i++){
            distributed[i] = a;
        }
        for(int i=0;i<rem;i++){
            distributed[i] += 1;
        }
```

```
        for(int i=0;i<box;i++){
            System.out.printf("Box %d : %d\n",i+1,distributed[i]);
        }
    }
}
```

Output

10

4

Box 1 : 3

Box 2 : 3

Box 3 : 2

Box 4 : 2

105.sum of min and max element

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int min = Integer.MAX_VALUE;
        int max = Integer.MIN_VALUE;
        for(int i=0;i<n;i++){
            if(arr[i]<min){
                min = arr[i];
            }
            if(arr[i]>max){
                max = arr[i];
            }
        }
        System.out.print(min+max);
    }
```

```
}
```
Output
5
10 20 40 50 30
60

106.pythagorean triplet

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int a = input.nextInt();
        int b = input.nextInt();
        int c = input.nextInt();
        int max = Math.max(a,Math.max(b,c));
        int x,y;
        if(max == a){
            x = b;
            y = c;
        }
        else if(max == b){
            x = a;
            y = c;
        }
        else{
            x = a;
            y = b;
        }
        if(x * x + y * y == max * max){
            System.out.print("Pythagorean triplet");
        }
        else{
            System.out.print("Not Pythagorean triplet");
        }
```

```
    }
}
```

107.array element difference

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n;i++){
            int sub = 0;
            if(i==0){
                System.out.print(arr[i]+" ");
            }
            else{
                sub = arr[i] - arr[i-1];
                System.out.print(sub+" ");
            }
        }
    }
}
```

Output

```
6
4 8 15 16 23 42
4 4 7 1 7 19
```

108.missing array finder

```
import java.util.*;
public class Main{
    public static void main(String[] args){
```

```
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        Arrays.sort(arr);
        int max = arr[n-1];
        int min = arr[0];
        boolean found = false;
        for(int i=min;i<=max;i++){
            boolean present = false;
            for(int j=0;j<n;j++){
                if(arr[j]==i){
                    present = true;
                    break;
                }
            }
            if(!present){
                System.out.print(i);
                found = true;
                break;
            }
        }
        if(!found){
            System.out.print(0);
        }
    }
}
```

Output

5

0 1 3 4 5

2

109.valid IP address or not

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        String[] tokens = str.trim().split("\\.");
        int n = tokens.length;
        if(n!=4){
            System.out.print("Invalid IP Address");
            return;
        }
        for(int i=0;i<n;i++){
            String word = tokens[i];
            int len = word.length();
            if(!word.matches("\\d+")){
                System.out.print("Invalid IP Address");
                return;
            }
            if(len>1 && word.startsWith("0")){
                System.out.print("Invalid IP Address");
                return;
            }
            int num = Integer.parseInt(word);
            if(num<0 || num>255){
                System.out.print("Invalid IP Address");
                return;
            }
        }
        System.out.print("Valid IP Address");
    }
}
```

Output

255.10.9.1

Valid IP Address

110.Write a program to swap characters at even and odd positions in a string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        char[] arr = str.toCharArray();
        int n = arr.length;
        for(int i=0;i<n-1;i+=2){
            char temp = arr[i];
            arr[i] = arr[i+1];
            arr[i+1] = temp;
        }
        String result = new String(arr);
        System.out.print(result);
    }
}
```

Output

apple

Palpe

111.swap nearby value in array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        for(int i=0;i<n-1;i+=2){
            int temp = arr[i];
            arr[i] = arr[i+1];
```

```
            arr[i+1] = temp;
        }
        for(int i : arr){
            System.out.print(i+" ");
        }
    }
}
```

Output

6

1 2 3 4 5 6

2 1 4 3 6 5

112.temperature should be greater than previous and next day

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int count=0;
        for(int i=1;i<n-1;i++){
            if(arr[i]>arr[i-1] && arr[i]>arr[i+1]){
                count++;
            }
        }
        System.out.print(count);
    }
}
```

Output

5

32 33 31 35 32

2

113.minimum number of boxes that can hold given apples

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int bunches = input.nextInt();
        int boxes = input.nextInt();
        int[] apples = new int[bunches];
        int[] capacities = new int[boxes];
        int count = 0,totalApples = 0;
        int filled = 0;
        for(int i=0;i<bunches;i++){
            apples[i] = input.nextInt();
        }
        for(int i=0;i<boxes;i++){
            capacities[i] = input.nextInt();
        }
        for(int j=0;j<bunches;j++){
            totalApples += apples[j];
        }
        Arrays.sort(capacities);
        for(int k=boxes-1;k>=0;k--){
            filled += capacities[k];
            count++;
            if(filled>=totalApples){
                System.out.print(count);
                break;
            }
        }
    }
}
```

Output

4

5

2 4 3 1
3 3 2 1 4
3

114.anagram with considering space

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s1 = input.nextLine();
        String s2 = input.nextLine();
        int len1 = s1.length();
        int len2 = s2.length();
        if(len1 != len2){
            System.out.print("Not Anagram");
            return;
        }
        char[] arr1 = s1.toCharArray();
        char[] arr2 = s2.toCharArray();
        Arrays.sort(arr1);
        Arrays.sort(arr2);
        if(Arrays.equals(arr1,arr2)){
            System.out.print("Anagram");
        }
        else{
            System.out.print("Not Anagram");
        }
    }
}
```

Output
the listen
silentthe
Not Anagram

115.anagram without considering space

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s1 = input.nextLine();
        String s2 = input.nextLine();
        s1 = s1.replaceAll(" ","").toLowerCase();
        s2 = s2.replaceAll(" ","").toLowerCase();
        if(s1.length() != s2.length()){
            System.out.print("Not Anagram");
            return;
        }
        char[] arr1 = s1.toCharArray();
        char[] arr2 = s2.toCharArray();
        Arrays.sort(arr1);
        Arrays.sort(arr2);
        if(Arrays.equals(arr1,arr2)){
            System.out.print("Anagram");
        }
        else{
            System.out.print("Not Anagram");
        }
    }
}
```

Output

silenteht

the listen

Anagram

116.longest common prefix

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s1 = input.nextLine();
```

```
        String s2 = input.nextLine();
        StringBuilder sb = new StringBuilder();
        int minlen = Math.min(s1.length(),s2.length());
        for(int i=0;i<minlen;i++){
            if(s1.charAt(i)==s2.charAt(i)){
                sb.append(s1.charAt(i));
            }
            else{
                break;
            }
        }
        System.out.print(sb.toString());
    }
}
```

Output

flower

flowerpot

flower

117.bull and cow

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String secret = input.nextLine();
        String guess = input.nextLine();
        int bull=0,cow=0;
        StringBuilder sRem = new StringBuilder();
        StringBuilder gRem = new StringBuilder();
        for(int i=0;i<secret.length();i++){
            char s = secret.charAt(i);
            char g = guess.charAt(i);
            if(s==g){
                bull++;
            }
```

```
            else{
                sRem.append(s);
                gRem.append(g);
            }
        }
        for(int i=0;i<gRem.length();i++){
            char c = gRem.charAt(i);
            int index = sRem.indexOf(String.valueOf(c));
            if(index!=-1){
                cow++;
                sRem.deleteCharAt(index);
            }
        }
        System.out.printf("%dA%dB",bull,cow);
    }
}
```

Output

1807

7810

1A3B

118.number of palindromic numbers in an array

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = input.nextInt();
        }
        int num,temp;
        int count=0,digit,rev=0;
        for(int i=0;i<n;i++){
            num = arr[i];
```

```
            temp = num;
            rev=0;
            while(temp>0){
                digit = temp%10;
                rev = rev*10+digit;
                temp/=10;
            }
            if(rev==num){
                count++;
            }
        }
        System.out.print(count);
    }
}
```

Output

6

121 345 11 987 88 202

4

119.Group anagrams

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        int n = input.nextInt();
        input.nextLine();
        String[] str = new String[n];
        for(int i=0;i<n;i++){
            str[i] = input.nextLine().toLowerCase();
        }
        boolean[] visited = new boolean[n];
        for(int i=0;i<n;i++){
            if(visited[i]) continue;
            System.out.print(str[i]);
            visited[i] = true;
```

```
            for(int j=i+1;j<n;j++){
                if(visited[j]) continue;
                if(str[i].length()==str[j].length()){
                    char[] a = str[i].toCharArray();
                    char[] b = str[j].toCharArray();
                    Arrays.sort(a);
                    Arrays.sort(b);
                    boolean isAnagram = true;
                    for(int k=0;k<a.length;k++){
                        if(a[k]!=b[k]){
                            isAnagram = false;
                            break;
                        }
                    }
                    if(isAnagram){
                        System.out.print(" "+str[j]);
                        visited[j] = true;
                    }
                }
            }
            System.out.println();
        }
    }
}
```

Output

5

Cat

act

rat

art

car

cat act

rat art

car

120.remove all adjacent duplicates in string

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        char[] stack = new char[n];
        int top=-1;
        for(int i=0;i<n;i++){
            char c = str.charAt(i);
            if(top>=0 && stack[top] == c){
                top--;
            }
            else{
                top++;
                stack[top] = c;
            }
        }
        for(int i=0;i<=top;i++){
            System.out.print(stack[i]);
        }
    }
}
```

Another method

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        StringBuilder sb = new StringBuilder(s);
        int i=0;
        while(i<sb.length()){
            if(i>0 && sb.charAt(i) == sb.charAt(i-1)){
                sb.deleteCharAt(i);
```

```
            sb.deleteCharAt(i-1);
            i-=2;
        }
        i++;
    }
    System.out.print(sb.toString());
  }
}
```

Output

abbaca

ca

azxxzy

ay

121.find the index of the first occurrence in the string

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner input = new Scanner(System.in);
    String haystack = input.nextLine();
    String needle = input.nextLine();
    if(haystack.contains(needle)){
      System.out.print(haystack.indexOf(needle));
    }
    else{
      System.out.print("-1");
    }
  }
}
```

Output

sadbutsad

sad

0

leetcode

leeto

-1

122.valid parentheses

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        int n = str.length();
        int top = -1;
        char[] stack = new char[n];
        boolean isValid = true;
        for(int i=0;i<n;i++){
            char c = str.charAt(i);
            if(c=='(' || c=='{' || c=='['){
                stack[++top] = c;
            }
            else{
                if(top==-1){
                    isValid = false;
                    break;
                }
                char open = stack[top--];
                if((c==')' && open!='(') || (c=='}' && open!='{') || (c==']' &&
open!='[')){
                    isValid = false;
                    break;
                }
            }
        }
        if(top!=-1){
            isValid = false;
        }
        System.out.print(isValid);
    }
```

```
}
```

Output

{[(})]
false

123.remove stars from a string and closest non-star character to its left

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str = input.nextLine();
        StringBuilder sb = new StringBuilder();
        for(int i=0;i<str.length();i++){
            sb.append(str.charAt(i));
        }
        int i=0;
        while(i<sb.length()){
            char c = sb.charAt(i);
            if(c == '*'){
                if(i>0){
                    sb.deleteCharAt(i);
                    sb.deleteCharAt(i-1);
                    i=i-1;
                }
                else{
                    sb.deleteCharAt(i);
                }
            }
            else{
                i++;
            }
        }
        System.out.print(sb.toString());
    }
}
```

Output
leet**cod*e
lecoe

124.backspace string compare

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String str1 = input.nextLine();
        String str2 = input.nextLine();
        StringBuilder sb1 = new StringBuilder(str1);
        StringBuilder sb2 = new StringBuilder(str2);
        int i=0;
        while(i<sb1.length()){
            char c = sb1.charAt(i);
            if(c == '#'){
                if(i>0){
                    sb1.deleteCharAt(i);
                    sb1.deleteCharAt(i-1);
                    i=i-1;
                }
                else{
                    sb1.deleteCharAt(i);
                }
            }
            else{
                i++;
            }
        }
        int j=0;
        while(j<sb2.length()){
            char c = sb2.charAt(j);
            if(c == '#'){
                if(j>0){
```

```
                sb2.deleteCharAt(j);
                sb2.deleteCharAt(j-1);
                j=j-1;
            }
            else{
                sb2.deleteCharAt(j);
            }
        }
        else{
            j++;
        }
    }
    if(sb1.toString().equals(sb2.toString())){
        System.out.print("True");
    }
    else{
        System.out.print("False");
    }
  }
}
```

Output

ab##

c#d#

True

125.reverse string II

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner input = new Scanner(System.in);
    String str = input.nextLine();
    int k = input.nextInt();
    char[] arr = str.toCharArray();
    for(int start=0;start<arr.length;start+= 2*k){
        int i=start;
```

```
            int j=Math.min(start + k - 1,arr.length - 1);
            while(i<j){
                char temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
                i++;
                j--;
            }
        }
        System.out.print(new String(arr));
    }
}
```

Output

abcdefg

2

bacdfeg

126.isomorphic substring

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        String s1 = input.nextLine();
        String s2 = input.nextLine();
        int[] arr1 = new int[256];
        int[] arr2 = new int[256];
        boolean isIsomorphic = true;
        if(s1.length()!=s2.length()){
            isIsomorphic = false;
        }
        else{
            for(int i=0;i<s1.length();i++){
                if(arr1[s1.charAt(i)]!=arr2[s2.charAt(i)]){
                    isIsomorphic = false;
                    break;
```

```
            }
            arr1[s1.charAt(i)]++;
            arr2[s2.charAt(i)]++;
        }
    }
    if(isIsomorphic){
        System.out.print(isIsomorphic);
    }
    else{
        System.out.print(isIsomorphic);
    }
  }
}
```

Output
1.egg
add
true
2.foo
bar
false

127.pangram

```
import java.util.*;
public class Main{
  public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    String str = sc.nextLine();
    int n = str.length();
    String s = str.toLowerCase();
    int[] freq = new int[256];
    boolean isPangram = true;
    for(int i=0;i<n;i++){
        freq[s.charAt(i)]++;
    }
```

```
        for(char c='a';c<'z';c++){
            if(freq[c]==0){
                isPangram = false;
                break;
            }
        }
        if(isPangram){
            System.out.print("Pangram");
        }
        else{
            System.out.print("Not Pangram");
        }
    }
}
```

Output

The quick brown fox jumps over the lazy dog

Pangram

hello world

Not Pangram

128.longest common prefix in array of strings

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        sc.nextLine();
        String[] words = new String[n];
        for(int i=0;i<n;i++){
            words[i] = sc.nextLine();
        }
        String prefix = words[0];
        for(int j=1;j<n;j++){
            String current = words[j];
            int k=0;
```

```
            while(k<prefix.length() && k<current.length() &&
prefix.charAt(k)==current.charAt(k)){
                k++;
            }
            prefix = prefix.substring(0,k);
            if(prefix.isEmpty()){
                break;
            }
        }
        System.out.print(prefix);
    }
}
```

Output

5
flower
flowerpot
flowing
flowerbasket
flows
flow

129.lemonade change

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] bills = new int[n];
        for(int i=0;i<n;i++){
            bills[i] = sc.nextInt();
        }
        boolean possible = true;
        int five=0,ten=0;
        for(int i=0;i<n;i++){
            int bill=bills[i];
```

```
            if(bill==5){
                five++;
            }
            else if(bill==10){
                if(five==0){
                    possible=false;
                    break;
                }
                five--;
                ten++;
            }
            else if(bill==20){
                if(ten>0 && five>0){
                    ten--;
                    five--;
                }
                else if(five>=3){
                    five-=3;
                }
                else{
                    possible=false;
                    break;
                }
            }
        }
        System.out.print(possible);
    }
}
```

Output

5

5 5 10 10 20

false

5

5 5 5 10 20

true

130.print the first occurrence of duplicate elements

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i]=sc.nextInt();
        }
        for(int i=0;i<n;i++){
            boolean isDuplicate=false;
            for(int j=0;j<i;j++){
                if(arr[i]==arr[j]){
                    isDuplicate=true;
                    break;
                }
            }
            if(!isDuplicate){
                System.out.print(arr[i]);
                if(i<n-1){
                    System.out.print(" ");
                }
            }
        }
    }
}
```

Output

6

10 22 10 20 11 22

10 22 20 11

131.3sum leetcode

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] = sc.nextInt();
        }
        Arrays.sort(arr);
        for(int i=0;i<n-2;i++){
            if(i>0 && arr[i]==arr[i-1]) continue;
            int left = i+1;
            int right = n-1;
            while(left<right){
                int sum=arr[i]+arr[left]+arr[right];
                if(sum==0){
                    System.out.println(arr[i]+" "+arr[left]+" "+arr[right]);
                    while(left<right && arr[left]==arr[left+1]) left++;
                    while(left<right && arr[right]==arr[right-1]) right--;
                    left++;
                    right--;
                }
                else if(sum<0){
                    left++;
                }
                else{
                    right--;
                }
            }
        }
    }
}
```

Output
6
-1 0 1 2 -1 -4
-1 -1 2
-1 0 1

132.email privacy

```
import java.util.*;
public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        int n = s.length();
        int count=0;
        int index = s.indexOf('@');
        for(int i=1;i<index;i++){
            count++;
        }
        if(count>3){
            System.out.print(s.charAt(0));
            for(int i=1;i<4;i++){
                System.out.print("*");
            }
            for(int i=4;i<n;i++){
                System.out.print(s.charAt(i));
            }
        }
        else{
            System.out.print(s.charAt(0));
            for(int i=1;i<3;i++){
                System.out.print("*");
            }
            for(int i=3;i<n;i++){
                System.out.print(s.charAt(i));
            }
```

```
        }
    }
}
```

Output

```
user@gmail.com
u**r@gmail.com
alice@example.com
a***e@example.com
johndoe@example.com
j***doe@example.com
```